GS1 DataMatrix UDI Label Management System — Master Plan

Project Overview

A full-stack web application for creating, managing, previewing, and printing GS1 DataMatrix UDI labels for medical devices. Compliant with **FDA 21 CFR 801.45** and **EU MDR 2017/745 Article 27**.

Key Business Rules

Two Label Modes

LOT MODE:

- Encodes: GTIN (01) + Lot Number (10) + Manufacturing Date (11)
- NO expiration date anywhere — not in the barcode, not in the HRI text, not in the form
- HRI display order: (01) → (10) → (11)

SERIAL MODE:

- Encodes: GTIN (01) + Serial Number (21) + Manufacturing Date (11) + Expiration Date (17)
- Expiration date included in barcode, HRI text, and form
- HRI display order: (01) → (21) → (11) → (17)

Encoding order vs Display order:

- DataMatrix ENCODING: fixed-length AIs first → (01)(11)(10) for lot, (01)(17)(11)(21) for serial
- HRI DISPLAY: identifier right after GTIN → (01)(10)(11) for lot, (01)(21)(11)(17) for serial

Tech Stack

- **Frontend:** React 18+ / TypeScript / Vite / Tailwind CSS
- **Backend:** Node.js / Express / TypeScript
- **Storage:** JSON files on disk (max 6 users, no SQL database)
- **Barcode:** bwip-js (gs1datamatrix symbology)
- **ZPL Generation:** Custom service converting templates + data → ZPL II commands
- **Printer:** Zebra GK420D via USB using Zebra Browser Print (client-side)
- **Label Preview:** Labelary API for ZPL → PNG rendering
- **PDF Export:** pdfkit

- **Batch PNG Export:** archiver (ZIP)
- **Date Math:** date-fns
- **File Locking:** proper-lockfile
- **Auth:** JWT (HTTP-only cookies) + bcryptjs

Architecture

- App runs on the PC with the USB printer
- LAN users access via `http://[server-IP]:3001`
- Server binds to `0.0.0.0` for LAN access
- LAN print queue: remote users queue jobs → printer PC operator clicks "Print Now"
- Zebra Browser Print only runs on the printer PC (localhost restriction)
- All data stored in `/data/*.json` files

Features

1. **Label Data Input** — GTIN validation, lot/serial toggle, date pickers, live DataMatrix preview with real GS1 encoding
2. **Automatic Expiration** — Configurable shelf life interval (years/months/days), auto-calculates from manufacturing date (serial mode only)
3. **Label Template Designer** — react-konva WYSIWYG editor, mm/in unit toggle, variable label sizes (square/rectangle), element positioning with boundary warnings, snap-to-grid, arrow key nudging
4. **Template Import/Export** — .gslabel and .gslabel-pack file formats, drag-and-drop import, conflict resolution
5. **ZPL Generation** — Template + data → ZPL II commands, Labelary preview
6. **PNG/PDF Export** — Single label and batch export (up to 500 labels), multi-page PDF
7. **Printer Connection** — Zebra Browser Print integration (USB), Network TCP driver (extensible)
8. **Sequential/Batch Printing** — Serial number pattern builder (prefix + range + suffix), progress tracking, pause/cancel
9. **LAN Print Queue** — Jobs from LAN users queue for the printer PC operator
10. **Product Profiles** — Save device configurations for reuse
11. **Auth & Roles** — Admin/operator roles, max 6 users, JWT sessions
12. **Audit Trail** — Append-only log of all actions for regulatory compliance

Agent Breakdown

Agent	Phases	Responsibility
Agent 1	1–3	Project setup, JSON storage, auth, GS1 encoding, data input form, live preview
Agent 2	4–5	Label template designer (react-konva), import/export, ZPL generation, PNG/PDF export
Agent 3	6–8	Printer module, batch printing, LAN print queue, audit UI, deployment
Agent 4	QA	Frontend testing, UX, label accuracy, HRI order verification
Agent 5	QA	Backend testing, API correctness, GS1 encoding, ZPL validation, data integrity

Execution Order

1. **Agent 1** → test checklist → confirm all passing
2. **Agent 2** → test checklist → verify ZPL on labelary.com
3. **Agent 3** → test checklist → physical printer test
4. **Agent 4** (QA) → bug report
5. **Agent 5** (QA) → bug report
6. Fix bugs from QA → re-run QA agents

Key Reference

- GS1 DataMatrix encoding reference: https://barcodes.pro/generator/gs1dmtx
- ZPL testing: http://labelary.com/viewer.html
- Zebra Browser Print: https://www.zebra.com/us/en/support-downloads/printer-software/browser-print.html
- Printer: Zebra GK420D, 203 DPI, USB connection
- Label example (LOT mode): GTIN 00850008393006, Lot MG96CI-260116, Mfg 260116, NO expiration
- Serial number format example: BWIII2026-1234 (letters, digits, hyphens)